

Programme	BSc Mathematics Honours			
Course Code	MAT4CJ204			
Course Title	BASIC LINEAR ALGEBRA			
Type of Course	Major			
Semester	IV			
Academic Level	200 – 299			
Course Details	Credit	Lecture/Tutorial per week	Practicum per week	Total Hours
	4	4	--	60
Pre-requisites	1.Familiarity with system of equations and their solutions 2. Knowledge about matrices and matrix operations.			
Course Summary	This course is a quick review of linear algebra, intended for students who have already taken a previous course in linear algebra or have some experience with vectors and matrices. It begins with the concepts of vector spaces, subspaces, bases and dimension. Linear transformations are introduced as ‘natural maps’ between vector spaces. The course opens up the classical finite dimensional inner product theory for the canonical reduction of a matrix as a special case of a self-adjoint operator.			

Course Outcomes:

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand and apply concepts related to vector spaces and subspaces, including determining whether a set forms a subspace and finding the span of a set	U	C	Internal Exam/Assignment/Seminar/Viva/ End Sem Exam
CO2	Demonstrate proficiency in analysing null spaces, column spaces, and linear transformations, including understanding the kernel and range of a linear transformation and contrasting the properties of null space and column space.	An	P	Internal Exam/Assignment/Seminar/Viva/ End Sem Exam
CO3	Evaluate and apply concepts related to bases, dimensionality, and rank of vector spaces, including understanding bases for null space and column space, determining dimensions of subspaces, and applying the rank theorem to systems of equations.	E	C	Internal Exam/Assignment/Seminar/Viva/ End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book	Linear Algebra and its Applications, Third Edition, David .C. Lay, Pearson Publications			
Module	Unit	Content	Hrs (48+ 12)	External Marks (70)
I	Module I			Min 15
	1	Section 4.1: Vector Spaces and Subspaces Vector Spaces and Subspaces, Subspaces, A Subspace Spanned by a Set.	14	
	2	Section 4.2: Null Spaces, Column Spaces, and Linear Transformations. The Null Space of a Matrix, An Explicit Description of Nul A.		
	3	Section 4.2: Null Spaces, Column Spaces, and Linear Transformations. The Column Space of a Matrix, The Contrast Between Nul A and Col A.		
	4	Section 4.2: Null Spaces, Column Spaces, and Linear Transformations. Kernel and Range of a Linear Transformation.		
	5	Section 4.3: Linearly Independent Sets; Bases. Linearly Independent Sets; Bases, The Spanning Set Theorem.		
	6	Section 4.3: Linearly Independent Sets; Bases. Bases for Nul A and Col A, Two Views of a Basis.		
II	Module II			Min 15
	7	Section 4.4: Coordinate Systems. Coordinate Systems, A Graphical Interpretation of Coordinates, Coordinates in $\mathbb{R}^n$.	12	
	8	Section 4.4: Coordinate Systems. The Coordinate Mapping.		
	9	Section 4.5: The Dimension of a Vector Space. The Dimension of a Vector Space.		
	10	Section 4.5: The Dimension of a Vector Space. Subspaces of a Finite-Dimensional Space, The Dimensions of Nul A and Col A.		
	11	Section 4.6: Rank Rank, The Row Space.		
	12	Section 4.6: Rank The Rank Theorem, Applications to Systems of Equations (Topics up to and including Example 5).		
III	Module III			Min 15
	13	Section 6.1: Inner Product, Length and Orthogonality The Inner Product, The Length of a Vector, Distance in $\mathbb{R}^n$.	12	
	14	Section 6.1: Inner Product, Length and Orthogonality Orthogonal Vectors, Orthogonal Complements, Angles in $\mathbb{R}^2$ and $\mathbb{R}^3$.		
	15	Section 6.2: Orthogonal Sets		

		Orthogonal Sets, An Orthogonal Projection (Topics up to and including Example 4).		
	16	Section 6.2: Orthogonal Sets Orthonormal Sets.		
	17	Section 6.4: The Gram-Schmidt Process The Gram -Schmidt Process, Orthonormal Bases.		
	18	Section 6.4: The Gram -Schmidt Process QR Factorization of Matrices.		
IV	Module IV			
	19	Section 7.1: Diagonalization of Symmetric Matrices Diagonalization of Symmetric Matrices.	10	Min 15
	20	Section 7.1: Diagonalization of Symmetric Matrices The Spectral Theorem. Spectral Decomposition.		
	21	Section 7.2: Quadratic Forms Quadratic Forms (Topics up to and including Example 3), Classifying Quadratic Forms.		
	22	Section 7.4: The Singular Value Decomposition The Singular Value Decomposition, The Singular Values of an $m \times n$ Matrix , The Singular Value Decomposition (Topics up to and including Example 4 only).		
V	OPEN ENDED		12	
Linear Algebra Lab Sessions Book: Mike Cohen, Practical Linear Algebra for Data Science, O'Reilly, 2019, ISBN 978-1-098-12061-0. Jupyter: https://github.com/mikexcohen/LinAlg4DataScience Choose lab demos and exercises for 12 hours as per lecturer's discretion. For Module I & II, Ch 2, 3, 5, 6 of book for Lab. For Module III, Ch 2 and Ch 9 of book for Lab. For Module IV, Ch 14 of book for Lab. Python and Jupyter review in Ch 16 of book.				
References				
1. Elementary Linear Algebra: Application Version,11/e, Howard Anton & Chris Rorres Wiley 2. Algebra Done Right, 3/e, Sheldon Axler, Springer Nature,2015. 3. Introduction to Linear Algebra, 6/e, Gilbert Strang, Wellesley-Cambridge Press. 4. Basic Linear Algebra, 2/e, T. S. Blyth and E.F. Robertson, Springer, 2002. 5. Linear Algebra, 2/e, Hoffman K and Kunze R, Prentice Hall of India,1991. 6. Bretscher, Otto. <i>Linear algebra with applications</i>. Vol. 52. Eaglewood Cliffs, NJ: Prentice Hall, 1997. 7. Blyth, Thomas Scott, and Edmund F. Robertson. <i>Basic linear algebra</i>. Springer Science & Business Media, 2013.				

***70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	1	2	2	3	0	3	0	3	0	0
CO 2	1	3	2	2	3	0	3	0	3	0	0
CO 3	3	2	3	3	3	0	3	0	3	0	0

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓